MIL-I-25992A(USAF)
10 September 1959
Superseding
MIL-I-25992(USAF)
27 February 1958

MILITARY SPECIFICATION

INDICATOR, BEARING-DISTANCE-HEADING ID-526A/ARN

1. SCOPE

1.1 This specification covers one type of aircraft navigation indicator, designated Indicator, Bearing-Distance-Heading ID-526A/ARN.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military

MIL-T-152	Treatment, Moisture- And Fungus-Resistant, Of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173	Varnish, Moisture-And Fungus-Resistant, For The Treatment Of Communications, Electronic, And Associated Electrical Equipment
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment (General Specification For)
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation For Delivery
MIL-T-9107	Test Reports, Preparation Of
MIL-T-21200	Test Equipment, For Use With Electronic and Fire Control Systems, General Specification For
MIL-L-25467	Lighting, Integral, Instrument, General Specification For
MIL-R-25717	Reliability Assurance Program For Electronic Equipment

MWM Corp Dayton
10-59 500

FSC 6605

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STANDARDS

Military

MS33556	Case, Aircraft Instrument, 3-1/4 Inch Size, Standard Dimensions For
MS33558	Numerals and Letters - Aircraft Instrument Dial, Standard Form Of

DRAWINGS

U.S. Air Force

55C12983	Face Design - Indicator, Bearing-Distance-Heading ID-526A/ARN
58D4054	Wiring Diagram - ID-526A/ARN Bearing-Distance-Heading Indicator, Schematic
58C4066	Installation Drawing - Indicator, Bearing Distance Heading

PUBLICATION

Air Force-Navy Aeronautical Bulletin

405	Storage Life - Aeronautical Articles

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 General Specification.- The requirements of MIL-E-5400 apply as requirements of this specification with the exceptions and additions, identified by similarity of paragraph headings called out herein. When the two documents conflict, this specification shall govern.

3.2.1 Classification.- The ID-526A/ARN indicator shall be class 2 as defined in MIL-E-5400.

3.2.2 Approval of Nonstandard Parts and Materials.- Nonstandard parts and materials approval shall be based upon the requirements specified in MIL-E-5400 for a category III equipment contract.

3.2.3 Cases and Front Panels.- The exterior finish of the metal case and metal front panel shall be durable dull black.

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3.2.4 Fungus Treatment.- Approval for fungus treatment of parts and materials shall be requested from the procuring activity. Where used, fungus treatment shall be in accordance with Specification MIL-T-152, using a fungistatic coating material conforming to Specification MIL-V-173, type II.

3.2.5 Life.- Reliable life requirements shall be in accordance with Specification MIL-R-25717 and this specification. When the two documents conflict, this specification shall govern.

3.2.5.1 Operating Life.- The ID-526A/ARN indicator shall have a minimum operating life of 5000 hours.

3.2.5.2 Service Life.- The equipment shall satisfy the operating life requirement of 3.2.5.1 when operated continuously or intermittently anytime during a minimum service life of 5 years. Service life shall include conditions of use, maintainability, environment, operation, storage, shipping, handling, etc.

3.2.5.3 Reliability.- The ID-526A/ARN indicator shall have a mean-time-to-failure of at least 400 hours.

3.2.5.4 Failures

3.2.5.4.1 Nonpredictable Failures.- Nonpredictable failures of those parts whose average life exceed the life requirements of this equipment (transistors, capacitors, resistors, etc.) shall be limited as specified in 3.2.5.3.

3.2.5.4.2 Wear-Out Failures.- Parts requiring replacement because of their usage or the present state of the art (bearings, etc.) shall be specified by the contractor with their recommended replacement intervals and subject to the approval of the procuring activity.

3.2.6 Maintenance Provisions

3.2.6.1 Operational Checkout Provisions.- When required, special or unique items of test equipment shall be developed in accordance with Specification MIL-T-21200 and shall be provided by the contractor.

3.2.6.2 Maintainability Provisions.- Consistent with other requirements specified herein, the design of the ID-526A/ARN indicator shall permit ease and speed of maintainability for both preventive and corrective maintenance with the equipment either installed or on the test bench under environmental conditions of service use.

3.2.7 Service Conditions (Electrical)

3.2.7.1 A-C Power Source.- The equipment shall operate on 108V to 121V, single phase, 380 to 420 cps ac.

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3.2.7.1.1 Lighting.- The lighted portion of the indicator shall operate on 5V ±0.1V ac, 2 volt-amperes.

3.2.7.2 D-C Power Source.- The indicator shall operate on 22V to 29V dc.

3.2.7.3 Power Ratings.- The nominal power required by the equipment shall not exceed the following:

a. A-C Power: 20 volt-amperes
b. D-C Power: 3 amperes

3.3 Design and Construction.- The ID-526A/ARN indicator shall be designed to present magnetic heading on a rotating card simultaneously with bearings to different targets or destinations and distance to the target or destination.

3.3.1 Indicator Face.- The design and arrangement of the indicator face shall be in accordance with Drawing 55C12983 and shall be subject to approval by the procuring activity prior to fabrication.

3.3.1.1 Rotating Card.- The rotating card of the indicator shall be driven by means of a motor and control transformer.

3.3.1.1.1 Azimuth Servo Amplifier.- The azimuth servo amplifiers, used to drive the azimuth card, shall be an integral part of the indicator, contained within the indicator case, but not necessarily within the hermetically sealed portion of the case. This amplifier shall not require electron tubes. The azimuth servo amplifier shall receive the output from equipments such as the Type J-4 compass transmitter with rotor excitation at 115V, 400 cps, and line-to-line rotor voltage of 11.8 volts open circuit.

3.3.1.2 Pointers.- The pointers shall be light as practicable, sufficiently rigid to prevent oscillation under vibration conditions, and firmly attached to the mechanism, yet readily adjustable.

3.3.1.2.1 The bi-material bending effect of the pointer under various temperature conditions specified herein shall be held to a minimum and shall not cause the pointers to touch each other, the dial, or the cover glass.

3.3.1.2.2 The length of the pointers from the center of the hub to the tip shall come up to, but not overlap, the shortest graduations; a tolerance of ±0.015 inch shall apply.

3.3.1.2.3 If counterbalances are required for the pointers, they shall not obscure the counter reading in any pointer position.

3.3.1.3 Request Heading Index.- The request heading index (or memory index) shall be light as practicable and sufficiently rigid to prevent oscillations under vibration conditions.

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3.3.1.3.1 The bi-material bending effect under various temperature conditions specified herein shall be held to a minimum and shall not cause the index to touch the azimuth card or the pointers.

3.3.1.3.2 The index shall rotate with the card to within ±0.25° under all service conditions specified herein, except when being set by the knob.

3.3.1.3.3 To preclude unbalancing the card index assembly, it may be necessary to counterbalance the index. If this is done, the counterbalance shall not be visible to the pilot.

3.3.1.3.4 The index shall be as close to the card as possible without dragging enough to deface the card with constant use. The index shall be placed behind the pointers and the fiducial marker and shall remain in set position under any service conditions.

3.3.1.4 Fiducial Markers.- The upper and lower fiducial markers shall be sufficiently rigid to prevent oscillation under vibration conditions.

3.3.1.4.1 The bi-material bending effect of the markers under various temperature conditions specified herein shall be held to a minimum and shall not cause the markers to touch the request heading index or the pointers.

3.3.1.4.2 The fiducial markers shall be as close to the azimuth card as possible, behind the pointers, and clear of the request heading index.

3.3.1.5 Reference Marks.- Reference marks shall be placed under the cover glass. With the azimuth card so set that North is under the fiducial marker, reference marks shall be placed at 45°, 90°, 135°, 225°, 270° and 315°.

3.3.1.6 Distance Display.- The distance display shall be a digital counter type presentation of nautical miles from 0000 to 9999. The units display shall have 1- and 1/2-mile index lines. The "thousands", "hundreds", "tens", and "units" numerals shall be individually driven by receiver synchros with a 1:1 ratio. The distance receiver synchros shall receive excitation from Kearfott Type RS911-6 or equivalent transmitter synchros. The "units" and "tens" and "hundreds" and "thousands" numerals shall be centered each 36° of synchro shaft rotation, with zero indication corresponding to electrical zero of the associated receiver synchro. Each receiver synchro set of leads shall terminate in five pins of the indicator connector.

3.3.1.7 Malfunction Warning Flag.- A three-position flag shall indicate non-operation, normal operation, and a malfunction in associated equipment.

3.3.1.7.1 Nonoperative Position.- When the flag is not energized, it shall indicate "OFF".

3.3.1.7.2 Normal Operation Position.- When in normal operation, the flag shall indicate a blank surface blending in with the background.

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3.3.1.7.3 Malfunction Position.- A malfunction in associated equipment or other specified conditions, such as excessive range shall actuate a 28V d-c signal to the flag mechanism, causing display of the "LIM" position.

3.3.2 Indicator Case.- The indicator case shall conform to Standard MS33556; the "L" length of which shall not exceed 6-7/8 inches. The internal mechanism of the indicator case shall be so designed that it may be removed from the case, replaced, and the case resealed without the use of special tools not specifically approved by the procuring activity.

3.3.2.1 Hermetic Seal.- The indicator case shall be hermetically sealed. The hermetic seal design shall be impervious to any atmospheric condition encountered by the indicator.

3.3.2.1.1 Filling Medium.- The case shall be filled with at least 88 percent nitrogen and the remainder helium. The filling medium shall be at least 98 percent pure, free of dust particles and shall contain no more than 0.006 milligram of water vapor per liter (dewpoint -65°C (-85°F) at the filling pressure.

3.3.2.1.2 Filling Pressure.- The filling pressure shall be at least 15.7 psi absolute. The allowable leak rate for each indicator shall not be more than 1.5 micron cubic foot per hour measured with a pressure differential of 1 atmosphere.

3.3.2.2 Cover Glass.- The cover glass shall be suitable for hermetic sealing (with internal gas pressure of 15.7 psi absolute) to an altitude of 70,000 feet and shall be clear, flat, and free from flaws. The distance between the inside of the cover glass and the azimuth card shall be the minimum practical.

3.3.2.2.1 An aircraft outline shall be so placed on the underside of the glass cover that it will appear white against a black background.

3.3.2.3 Lighting.- The indicator case shall be integrally lighted in accordance with Specification MIL-L-25467. Light shall be so evenly distributed, without hot spots of light intensity, that all parts required for indicator operation are readable under all ambient light conditions. If any shields or masks are used on the indicator face, that shall not interfere with the presentation visibility.

3.3.2.3.1 Illumination.- For illumination, type 337 SR or NR and type CM 680 lamps may be used in conjunction with red filtering material.

3.3.2.3.2 Lamp Reliability.- The lamps shall operate at least 2000 hours without replacement under the specified service conditions.

3.3.2.3.3. Wiring.- Lighting wiring shall be connected to separate pins in the receptacle.

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3.3.2.4 Heading Index Knob.- The heading index knob shall be located in the lower right corner (when facing) of the indicator case.

3.3.2.4.1 Knob Positions

a. In normal operation the knob shall set the request heading index on the azimuth card. Clockwise rotation of the knob shall effect clockwise rotation of the index. One revolution of the knob shall move the index approximately 40°. Index rotation shall not affect the azimuth card position.

b. When the position of the index is given to the autopilot as a command signal, the knob shall be pulled out to a second position. In the pulled-out or second position, knob rotation shall not affect the indicator presentation or output signal. Excursion of the pullout shall not be less than 1/2 inch. Positive indication of a stop or detent shall be felt in both positions of the knob.

c. If the autopilot is disconnected by the pilot's normal operating autopilot switch, the index knob shall return to the push-in position and shall not control the indicator output to the autopilot.

3.3.2.4.2 The contractor shall provide means for locking the knob in the pull-out position by fractional rotation or similar one-hand operation methods; however this means shall not prevent the normal autopilot controls from overriding the knob control and shall allow the knob to snap back to the push-in position.

3.3.2.4.3 Autopilot Transmitter.- The automatic flight control subsystem transmitter shall be a Clifton Precision type CGC-10-AS6, or equal. The transmitter shall have 5 separate leads terminating in separate pins of the electrical connector. The transmitter leads shall be shielded if necessary.

3.3.2.5 Function Inserts.- Inserts in the face plate shall indicate the functions of the Number 1 and Number 2 pointers, respectively, and the distance counter pointer relationship. The inserts with the proper coding shall be specified in each contract prior to acceptance by the procuring activity. Inserts shall be easily attached or removed from the face plate while the indicator is installed on the instrument panel (front mounting).

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3.3.2.5.1 Inserts to be installed, or supplied as spare parts, shall be selected from the following list:

PTR. NO. 1 RAD. COMP.	MILES TO NO. 1
PTR. NO. 1 VOR	MILES TO NO. 2
PTR. NO. 1 RHO THETA	NO DIST. IND.
PTR. NO. 1 ADF	BLANK (No printing)
PTR. NO. 2 RAD. COMP	
PTR. NO. 2 VOR	
PTR. NO. 2 TacAN	

The above inserts shall be dimensionally interchangeable with each other.

3.3.3 Circuit Design.- The circuit design of the indicator shall be in accordance with Drawing 58D4054.

3.3.4 Mounting.- The indicator shall be designed for front mounting. Each of the four mounting lugs of the indicator face shall withstand loads up to 150 pounds when attached to the mounting plate by the use of screws.

3.3.5 Connector.- The electrical connector or connectors used shall be Cannon type K miniature, or equal, and shall be designed for hermetic sealing if connected directly to the hermetically sealed part of the indicator.

3.4 Performance.- The Indicator, Bearing-Distance-Heading ID-526A/ARN shall operate as the primary display of the heading reference output of gyro magnetic compasses, such as the Types J-4 and N-1.

3.4.1 The indicator shall display:

a. Bearing and distance outputs of the AN/ARN-21 radio set with a pointer and counter.

b. Magnetic heading with accuracies suitable for use as the primary references in the cockpit.

c. Simultaneous bearings to two different reference points determined by outputs from navigation equipments (TacAN, AN/ASN-7( ), AN/ASN-15, and AN/ASN-24 computer sets, radio compass, etc.).

3.4.2 The indicator shall provide a synchro output to the automatic flight control subsystem of the aircraft as determined and set-in by the operator.

3.5 Weight.- The total weight of the ID-526A/ARN indicator shall not exceed 4 pounds.

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3.6 Operation Markings

3.6.1 Indicator Face.- The azimuth card numerals shall distinctly indicate the graduation to which each applies.

3.6.2 Indicator Markings.- All markings shall be durable to withstand the service conditions specified herein without visible deterioration. The indicator face shall be marked in accordance with Drawing 55C12983. The form of letters and numerals on the indicator shall conform to Standard MS33558 and the requirements specified herein.

3.6.2.1 Distance Display.- The numbers on the distance display counter shall increase from top to bottom, i.e., as the number moves down, the next number shall be smaller. Numerals shall be 3/16 inch high, minimum. When a number is centered, half of the coming and half of the going number of the units (1 mile) display shall be visible.

3.6.2.2 Function Inserts.- The lettering on the inserts shall be approximately 1/16 inch high, arranged as practicable. Lettering shall be lighted as part of the instrument lighting.

3.6.2.3 Visibility of Presentation.- The pointers, card numerals, at least half of the length of the card graduations, and the counter numbers shall be visible from any point within the frustum of a cone, the sides of which make an angle of 25° with a perpendicular to the dial, the small diameter being the cover glass of the indicator case.

3.6.2.4 Radioactive Materials.- Radioactive materials shall not be used in finishes for letters, numerals, lines, and pointers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 The inspection and testing of the Indicator, Bearing-Distance-Heading ID-526A/ARN shall conform to section 4 of Specification MIL-E-5400 with exceptions and additions called out herein. When the two documents conflict, this specification shall govern.

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4.2.1 The approval of the production equipment or sample equipments required in 4.6.1 shall be contingent upon the authorization by the procuring activity for the use of all parts and materials.

4.3 Acceptance Tests.- Acceptance tests shall consist of:

a. Individual tests
b. Sampling plans and tests

4.3.1 Individual Tests.- Each indicator shall be subjected to the following tests:

TEST	APPLICABLE PARAGRAPH
Examination of Product	4.5.1
Operational Tests	4.5.2
Reliability Test	4.5.5.1

4.3.2 Sampling Plans and Tests

4.3.2.1 Sampling Plan A.- One indicator shall be selected at random from every 25 or fraction thereof manufactured on the production run and subjected to the following tests.

TEST	APPLICABLE PARAGRAPH
Environmental Tests, as follows:	
High and Low Temperature Tests	4.5.3.1
Altitude Tests	4.5.3.3
Vibration	4.5.3.5
Magnetic Effect	4.5.4

4.3.2.1.1 Rejection and Retest.- When one indicator selected from a production run fails to meet this specification, no indicators still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.3.2.1.2 Individual Tests May Continue.- For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of indicators on hand or produced later shall not be made until it is determined that indicators meet all the requirements of the specification.

4.3.2.2 Sampling Plan B.- The first 7 indicators from the initial production run shall be subjected to the reliability test described in 4.5.5.2.

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4.3.2.3 Sampling Plan C.- The testing of 1 indicator from sampling plan B shall be continued in accordance with the life test requirements of 4.5.6.

4.3.3 Defects In Items Already Accepted.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Testing Requirements

4.4.1 Test Conditions.- Unless otherwise specified, the indicator shall be subjected to the tests under the test conditions specified in Specification MIL-E-5400, using applicable reference voltages as specified under 3.2.7.

4.4.2 Standard Test Transmitter.- The indicator shall be tested with a standard test transmitter, Kearfott Type RS911-6, or equal, high precision transmitting synchro. The rotor of the transmitter shall be settable, rotatable, and readable through 360° of rotation in steps of 0.1°. The standard test transmitter shall have minimum and known errors.

4.4.2.1 The leads of rotor "HOT" and "COLD" shall be designated at will.

4.4.2.2 One of the stator leads shall be selected, designated, labeled "Z" and tied to the "COLD" rotor lead.

4.4.2.3 A voltage of 115 volts, 400 cycles shall be applied to the "HOT" lead and the "COLD" and "Z" leads shall be grounded. The rotor or stator shall be rotated until the output of the remaining leads is at a null or minimum voltage.

4.4.2.4 The voltage between the "HOT" lead and the remaining stator leads strapped shall be measured. If this voltage is less than the excitation voltage of the rotor, the synchro is at "LOW NULL" or the 180° position. If the voltage is greater than the rotor excitation voltage, the synchro rotor is at "HIGH NULL" or the zero-degree position.

4.4.2.5 The rotor shall be placed at the zero-degree or HIGH NULL position. A voltmeter shall then be connected across one of the remaining stator leads and the ground, with the second lead open. When the synchro rotor is turned clockwise, the voltage will increase before decreasing if the "X" stator lead is connected. The voltage will decrease before increasing if the "Y" stator lead is connected. All the leads of the test synchro shall be labeled in accordance with the results of the above tests.

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4.4.3 Test Reports.- Unless otherwise specified by the procuring activity, the contractor shall furnish, to the procuring activity, a test report in accordance with Specification MIL-T-9107 giving the results of the acceptance and preproduction tests specified herein.

4.5 Test Methods

4.5.1 Examination of Product.- The indicator shall be visually inspected to verify that the materials, design and construction, necessary mechanical measurements, marking, and workmanship comply with the requirements of this specification.

4.5.2 Operational Tests

4.5.2.1 Electrical Zero and Rotational Test.- The test transmitter shall be connected to the two bearing synchros in turn, as follows:

Test Transmitter	No. 1 Pointer Test	No. 2 Pointer Test
Hot	20	25
Cold	21	26
X	17	22
Y	18	23
Z	19	24

The pins shall be numbered in accordance with figure 1.

4.5.2.1.1 When the transmitter needle and rotor are at 0° position, the numbers 1 and 2 pointers shall indicate 0° with the azimuth card set at North under the top fiducial marker.

4.5.2.1.2 When the test transmitter is rotated clockwise with increasing position indication, the pointers of the indicators shall rotate with increasing positive indication.

4.5.2.1.3 When the test transmitter rotor is rotated counterclockwise with decreasing positive indication, the pointers of the indicators shall rotate with a decreasing positive indication.

4.5.2.2 Bearing Pointers Scale and Index Errors.- The standard test transmitter shall be connected as indicated in 4.5.2.1 and turned successively to bearings of 30°, 60°, 90°, 120°, 150°, 180°, 210°, 240°, 270°, 300°, 330° and 360°. The bearing pointer readings shall be recorded. The azimuth cards on the three paralleled indicators shall be set at North under the top fiducial marker.

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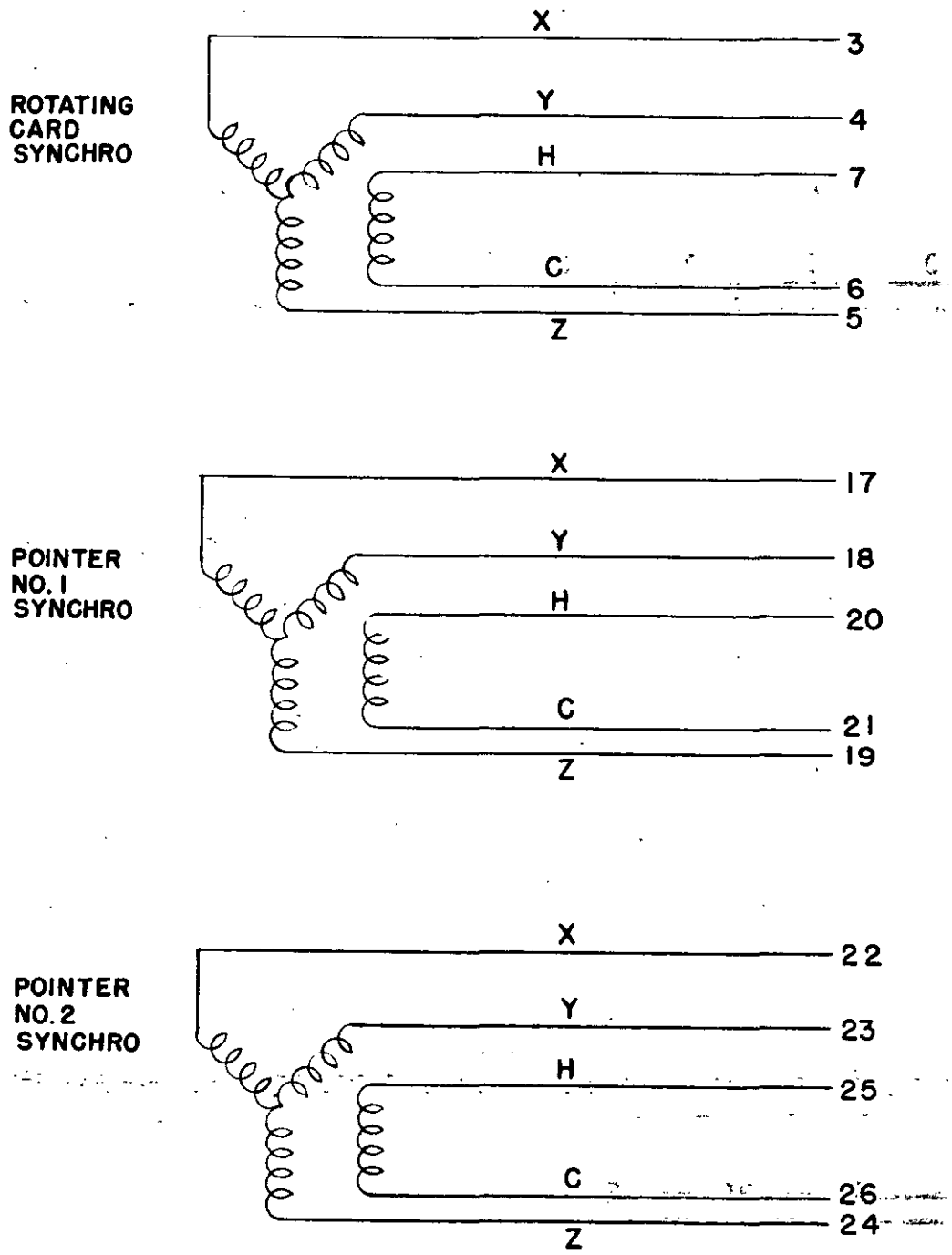


FIGURE 1. Pin Numbering

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4.5.2.2.1 The difference between the pointer bearings and the standard transmitter readings for each pointer successively shall be added algebraically and the algebraic sum divided by the number of bearings. The results, allowing for electrical zero, shall be the index error and shall not exceed ±0.5°. Any one reading shall not differ from the transmitter setting by more than ±1° after tapping.

4.5.2.3 Bearing Pointers Friction Errors.- The scale and index error tests (4.5.2.2) shall be repeated, except that the readings of the two bearing pointers of each of the three paralleled indicators shall be taken before tapping and after tapping on each heading.

4.5.2.3.1 The change in readings as a result of tapping shall be the friction errors, and shall not exceed 1.5° on each pointer.

4.5.2.3.2 This test may be combined with the scale and index error tests.

4.5.2.4 Bearing Pointers Damping.- Each bearing pointer synchro shall be connected in parallel to two other bearing synchros, or equivalent electro-mechanical impedances and excited from a transmitter synchro whose rotor is aligned 120° to the respective receiver synchro rotors of the bearing pointers.

4.5.2.4.1 The time required for oscillation of each bearing pointer to cease shall not exceed 4 seconds.

4.5.2.4.2 There shall be no evidence of motoring of the receiver synchros.

4.5.2.5 Azimuth Card Scale Error and Index Error

4.5.2.5.1 The test transmitter, connected to the input of the azimuth card drive, shall be turned counterclockwise successively to bearings of 30°, 60°, 90°, 120°, 150°, 180°, 210°, 240°, 270°, 300°, 330° and 360° and the azimuth card readings recorded.

4.5.2.5.1.1 The difference between the heading reading (card) and the transmitter setting for each successive reading shall be added algebraically and the algebraic sum divided by the number of readings. The result shall not exceed 0.25°.

4.5.2.5.2 The transmitter shall then be turned clockwise successively to 330°, 300°, 270°, 240°, 210°, 180°, 150°, 120°, 90°, 60°, 30° and 0° and the indicator card readings recorded.

4.5.2.5.2.1 The difference between the heading reading and the transmitter setting shall be added algebraically and the algebraic sum divided by the number of readings.

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4.5.2.5.3 The difference between the clockwise and the counterclockwise algebraic sum shall not exceed 0.25°. The scale error shall not exceed ±0.5°.

4.5.2.6 Distance Indicator Error.- A transmitter shall be connected to each receiver synchro distance indicator (units, tens, hundreds, thousands). An electro-mechanical impedance, equivalent to two equivalent paralled repeaters, shall be connected to each receiver. The transmitter synchros shall be set successively to 36°, 72°, 108°, 144°, 180°, 216°, 252°, 288°, 324° and 360°.

4.5.2.6.1 The corresponding drum or dial of each distance indicator shall not be different from the connected transmitter by more than 3° of the circumference of the drum or dial.

4.5.2.7 Request Heading Error.- With the azimuth card locked at 0°, the request heading index shall be set by the knob to successive readings of 30°, 60°, 90°, 120°, 150°, 180°, 210°, 240°, 270°, 300°, 330°, and 360° and the knob actuated for autopilot operation at each setting. The receiver synchro connected to the request heading transmitter shall rotate in the correct direction and shall be positioned within ±0.5° on all headings. The request heading transmitter shall be tested to assure the following characteristics of steering voltage signal to the autopilot:

Amplitude:	300 ±25 mv per degree
Null Signal Level:	Not to exceed 30 mv
Phase Shift:	Not to exceed 5°
Signal Symmetry:	Within 5 percent
Signal Linearity:	Linear within 5 percent to amplitude of 20°

4.5.2.8 Sealing.- Gas leakage shall be determined by a helium leak detector, using a suitable pressure differential (1 atmosphere) from outside to inside the case. Any leakage in excess of 1.5 micron cubic foot per hour shall be cause for rejection.

4.5.2.9 Lighting.- The indicator lighting shall be tested in accordance with Specification MIL-L-25467.

4.5.3 Environmental Tests.- The indicator shall be subjected to the following environmental tests in accordance with the indicated procedures of Specification MIL-E-5272 with exceptions and additions called out herein, to determine compliance with environmental requirements of Specification MIL-E-5400. All environmental tests resulting in destruction of the indicator shall be coordinated with and subject to approval of the procuring activity prior to actual testing.

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4.5.3.1 High Temperature and Low Temperature Tests, Procedure II.- The indicator shall be held at each temperature limit for 2 hours after stabilization. There shall be no evidence of sticking of any elements of the indicator during or after these tests.

4.5.3.1.1 The operating data obtained from these tests shall not exceed the allowable errors as specified in 4.5.2.2 through 4.5.2.7 by more than 0.5°.

4.5.3.2 Humidity Tests, Procedure I.- At the conclusion of this test the equipment shall be operated without malfunction of any display components.

4.5.3.3 Altitude Tests, Procedure VI.- The indicator shall be operated during this test in 10,000 foot increments to 70,000 feet without any reduction in precision, or evidence of leakage of gas in excess of 1.5 micron cubic foot per hour.

4.5.3.4 Salt Spray.- After testing, the indicator shall be examined and there shall be no evidence of external corrosion or deterioration which will effect subsequent operation.

4.5.3.5 Vibration Tests, Procedures IV and V

4.5.3.5.1 Vibration Error (Procedure IV).- With respective number 1 and number 2 bearing pointers connected in parallel, and other display elements properly energized, the indicator shall be vibrated in accordance with test procedure IV, the diameter of circular motion being 0.018 to 0.020 inch.

4.5.3.5.1.1 During the vibration period the pointer readings shall not differ from the transmitter setting by more than ±1°.

4.5.3.5.1.2 Azimuth cards shall show no appreciable difference in readings, and distance readings shall show no appreciable difference.

4.5.3.5.1.3 Repeated operation of the malfunction warning shall not show failure to operate.

4.5.3.5.1.4 The memory index marker shall not change relationship to the azimuth card by more than 0.1°.

4.5.3.5.1.5 The autopilot operation knob shall function with no failures in repeated operations.

4.5.3.6 Fungus Resistance Tests, Procedure I.- Upon completion of this test, the indicator shall be examined and operated to determine if any permanent damage incurred. Permanent damage shall be cause for rejection.

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4.5.4 Magnetic Effect.- The indicator, first not operating and then operating, on rated voltage shall be rotated in a vertical plane about a short bar magnetic compass with the nearest part of the indicator 5-1/2 inches from, and magnetically east or west of, the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225° and 315° from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in the normal upright position.

4.5.4.1 The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 2°.

4.5.5 Reliability Tests

4.5.5.1 Each equipment shall be subjected to a 7-hour functional reliability test, the procedure of which shall be prepared by the contractor and submitted to the procuring activity for approval. At least 98 percent of the equipments shall complete this test without failure.

4.5.5.1.1 Rejection and Retest.- When more than 1 indicator from a production run of 25 fail to pass the reliability test of 4.5.5.1, all equipments on the production run shall be withheld until full particulars concerning the cause, extent of failure, and corrective action taken are furnished to the Government inspector. All equipments that fail the specified test shall either be reworked to correct the defects and retested or replaced by an acceptable equipment prior to final acceptance of the entire production run.

4.5.5.2 A test of 200 operating hours shall be performed on equipments taken from the initial production run to indicate an adequate extended period of failure-free operation. Failures occurring before the completion of 200 operating hours shall be corrected and the test resumed.

4.5.6 Life Test.- The equipment (without vibration isolators) shall be subjected to a 2000-hour life test under the following simulated service conditions:

Temperature:	Room Ambient (25°C, +15°C or -5°C) (68°F to 104°F)
Altitude:	Normal (0 to 5000 feet)
Humidity:	Room Ambient (up to 90 percent)
Vibration:	Constant frequency, intermittent operation as specified in 4.5.6.1
AC Voltage:	115V, 400 cps and 5V
DC Voltage:	28V

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4.5.6.1 Vibration.- Once in each 24 hours of operation, the equipment shall be subjected to a 10-minute period of vibration. The vibration shall be constant frequency, within the range of 25 to 35 cps and total excursion of 0.015 to 0.030 inch, and need not be simple harmonic or purely linear motion. Equipment operation shall be observed during the 10-minute vibration period to determine if degradation has occurred due to microphonics, worn or damaged parts, charred insulation, etc.

4.5.6.1.1 After the 10-minute vibration period, the operational tests specified under 4.5.2 shall be performed. This cycle shall be repeated until the equipment has operated for the time specified herein.

4.5.6.2 A daily record of equipment performance, making particular note of any deficiencies, shall be maintained. In the event of parts failures during this test the defective parts shall be replaced and the operation resumed for the balance of the 2000-hour test period.

4.5.6.2.1 Failures shall be recorded throughout the test indicating the following:

a. Part
b. Part function
c. Nature of the failure
d. Operating environment at the time of failure
e. Number of hours the part operated prior to the failure

4.5.6.3 A report of all parts failures shall be submitted to the procuring activity for approval immediately upon completion of the first 200 hours and every 500 hours of operation thereafter. The reports shall include proposals of suitable and adequate design or material corrections for all failures which occurred.

4.5.6.4 Depending upon the nature and number of failures that occur, the life test may be stopped after 1000 hours of operation at the discretion and approval of the procuring activity.

4.6 Preproduction Testing

4.6.1 Preproduction Test Samples.- The contractor shall conduct tests on three preproduction sample indicators to determine that the design of the equipments meets the requirements specified herein.

4.6.2 Preproduction Tests.- Preproduction test shall consist of all tests described under 4.5, excluding 4.5.5 and 4.5.6, subject to the test conditions specified herein.

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5. PREPARATION FOR DELIVERY

5.1 All preservation, packaging, packing, marking, and testing shall conform to the requirements of Specification MIL-P-7936.

5.2 Marking of Shipment

5.2.1 Shipment Marking Nomenclature.- The shipment marking nomenclature shall be "INDICATOR, BEARING-DISTANCE-HEADING ID-526A/ARN".

5.2.2 Precautionary Marking.- The following precautionary marking shall appear on two opposite sides of each interior package, whenever practical, depending on the size of the carton and shall also appear on the exterior shipping container:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

5.2.3 Reinspection Date Markings.- The reinspection date markings shall be in accordance with ANA Bulletin 405.

6. NOTES

6.1 Intended Use.- The Indicator, Bearing-Distance-Heading ID-526A/ARN covered by this specification is intended for use as a standard indicator for navigational computers such as the AN/ASN-7A and as a primary heading reference replacing the Type V-7A or V-8 indicator and will function as a range and bearing indicator for operation with TacAN.

6.2 Ordering Data.- Procurement documents should specify the following:

a. That three indicators will be required as preproduction samples and that these preproduction samples will be subjected to the preproduction tests to determine compliance with the requirements of this specification.

b. The point of inspection for the preproduction tests.

c. Applicable requirements listed under "ordering data" of Specification MIL-P-7936.

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